INCH-POUND

MIL-D-48688A(AR)
20 JULY 1989
SUPERSEDING
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MILITARY SPECIFICATION

DISPENSER, GROUND: HE, M131 MINE SYSTEM LOADING, ASSEMBLING AND PACKING INCLUDING HE MINE, ANTIPERSONNEL AND ANTITANK

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification contains requirements not normally covered by the drawings, and provides quality assurance provisions for the loading, assembling and packing of Dispenser and Mine, Ground: Antipersonnel, Antitank HE, M131 System.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSMC N/A FSC 1345

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SPECIFICATIONS

MILITARY

MIL-Q-9858 - Quality Program Requirements.
MIL-S-45743 - Soldering, Manual Type High Reliability Electrical and Electronic Equipment.
MIL-S-46844 - Solder Bath Soldering of Printed Wiring Assemblies.
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions General Specification For.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWING

9292865 - Housing Assembly Out-of-Line Safety.
9332860 - Control and Sensor (Schematic).
9333036 - Control Indicator.
9337379 - Mine, Antitank: High Explosive.
9337380 - Spring Finger.
9337422 - Mine AP HE.
9337504 - Coil and Spacer Assembly.
9337505 - Loaded Tube Assembly M131 System.
9337506 - Dispenser and Mine, Ground M131 System.
9337507 - Loaded Module Assembly M131 System.
9337511 - Plate Assembly, Retainer.
9337525 - Cable and Connector Assembly.
9337533 - Wiring Diagram, MOPMS Dispenser.

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(Copies of specifications, standards, drawings, publications and other Government documents required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Material shall be in accordance with the applicable drawings, specifications, and standards.

3.2 Components and assemblies. The dispenser and components shall comply with all requirements specified on drawings 9333026, 93337506 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Requirements of dispenser and mine ground: antipersonnel/antitank M131. This section contains specific requirements pertaining to assembly and functioning of the Dispenser and Mine, Ground: Anti-Personnel/Antitank System.

3.3.1 Safe condition and circuit continuity verification.

3.3.1.1 Housing assembly, out-of-line safety. The Housing Assembly, Out-Of-Line Safety Device (Drawing 9292865) shall be in the full safe condition as evidenced by presence and proper position of the following safety features or components prior to insertion of control indicator into dispenser.

a. Interrupter- Present and in safe position

3.3.1.2 Dispenser activation circuit continuity/resistance. The Tube Drive Lines (Mine Battery Activation/Data Transfer Lines), Bellows Activator Line, Detonator Line and Magnetic Coupling Device Lines shall have the following circuit resistance when measured at the connector of the dispenser cable and connector assembly (Drawing 9337525).

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Line	Resistance
Tube Drive Line	.275 + .075 ohms
Bellows Actuator Line	.200 + .070 ohms
Detonator Line	.500 + .400 ohms
Magnetic Coupling Device Line	3.0 + 2.0 ohms

3.3.1.3 Soldering process control plan. All soldering shall be in accordance with MIL-S-45743 for hand soldering and MIL-S-46844 for machine (flow) soldering. All unique connections not identified in the applicable military specifications or drawings shall be highlighted in the contractor's soldering process control document in accordance with 4.5.

All solder training shall be conducted under the direction of personnel with Category C authority.

3.3.1.4 Polarity of coil, Magnetic Coupling Device (MCD)-9332919. The MCD coil will produce a magnetic field in accordance with dwg. 9332919 and will be tested as specified in 4.4.3.2.3.

3.3.1.5 Insulation resistance. The shields of the cable and connector assembly shall not be connected to the shell of connector P1 nor shall there be electrical continuity between the shields and the MCU bracket of the loaded module assembly 9292804.

3.3.1.6 Leak test. The dispenser shall not lose air pressure at a rate greater than .05 psi in one (1) minute when tested in accordance with 4.4.3.2.5.

3.3.2 Dispenser function.

3.3.2.1 Lamp "push-to-test" check. When B^+ is applied to the Control Indicator power connector the lamps on the control unit shall light when pressed.

3.3.2.2 Self test. When B^+ is applied to the Control Indicator power connector and the Control Indicator switch is turned to "test" the "Battery" lamp should light indicating that the battery voltage is correct and the "CKT OK" lamp should light indicating successful internal tests.

3.3.2.3 Data load. When the Control Indicator control switch is placed in the "Load" position, and after the proper data is transferred to the dispenser by the signal generator in accordance with dwg. 9333026, the "Data In" light should light indicating that data was entered into the Control Indicator.

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3.3.2.4 Dispenser arming. When the selector switch on the Control Indicator is pushed in and turned to the "ARM" position, the dispenser must arm and accept a proper dispense command to eject the stored mines.

3.3.2.5 Dispense command - RF. After arming, the dispenser shall accept a properly coded "DEPLOY" command of 9.0 ± 1.0 MV/M field strength sent by a signal generator. If the dispenser was not properly coded it must not accept a R.F. dispense command.

3.3.2.6 Dispense command - hardware. After arming, the dispenser shall accept a "DISPENSE" command from a M32 or M34 blasting machine or a M57 firing device when properly connected to the "HARDWIRE" terminals on the Control Indicator.

3.3.2.7 Coding of mines. If a group identification code is properly loaded into the dispenser control indicator, the identification code must be loaded into the mines after battery activation during the deployment cycle. "Mines coded with a group identification code and armed (enabled) must be able to have their self destruct time recycled or be "COMMAND DESTRUCTED" by the signal generator used to code that dispenser."

If the dispener control indicator does not receive a group identification code and receives a "DEPLOY" command, a "DUMMY" code will be loaded into the mines after battery activation in the deployment cycle. "Mines coded with a "DUMMY" code and armed cannot have their self destruct timer reset nor "COMMAND DESTRUCTED". The mines, loaded with the "DUMMY" code, shall have the capacity of functioning either prime fuze or destruct at time out of the preset self destruct time.

3.3.2.8 Mine field pattern. Mines ejected from the dispenser, given proper dispense command, must fall into an area defined as a half section of an annulus with a minor radius of 10 meters and a major radius of 40 meters.

3.4 Mine functions. Mines ejected from the dispenser, given proper dispense command and coded with a group identification code, shall be capable of functioning in the prime target sensor mode, a self-destruct mode, recycled self-destruct mode and command destruct mode. Mines given a "DUMMY" code will function only in a single self destruct function or in the prime target sensor mode.

3.5 Environmental conditions.

3.5.1 Operating temperature. The dispenser and mines shall perform as required over a temperature range of -35° to 145°F.

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3.5.2 Transportation vibration. The dispenser shall be safe to handle during transport and shall show no evidence of internal or external damage that will affect its function when tested specified in paragraph 4.3.2.2.1.

3.5.3 Seven foot drop. The dispenser shall be safe to handle and show no evidence of internal or external damage that will affect its function following a seven (7) foot drop test.

3.5.4 Hot temperature conditioning. The dispenser shall function properly when conditioned in accordance with 4.3.2.2.3.

3.5.5 Cold temperature conditioning. The dispenser shall function properly when conditioned in accordance with 4.3.2.2.4.

3.5.6 Ambient temperature conditioning. The dispenser shall function properly when conditioned in accordance with 4.3.2.2.5.

3.5.7 Dispenser functioning. The dispenser shall function properly when tested in accordance with 4.4.2.6.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough workmanlike manner; and all manufacturing, processing, and assembling operations shall be correctly performed. All parts and assemblies shall be clean, free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent(s). Exterior surface coating shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item.

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall be tested in accordance with and consist of the items and quantities listed in Table I.

4.3.2 Inspection and tests to be performed.

4.3.2.1 Components. See MIL-A-48078 and Table I specified herein.

4.3.2.2 Dispenser. The dispenser listed in Table I shall be subjected to the following tests:

4.3.2.2.1 Transportation Vibration (TV). In accordance with the requirement of MIL-STD-810, Method 514, Category 8. Test Procedure I Test Condition I.3.2.10.

4.3.2.2.2 Seven foot drop. Following (TV) the dispenser shall be safe to handle and show no evidence of internal or external damage that will affect its function following a seven (7) foot drop test.

4.3.2.2.3 Hot temperature conditioning. Following conditioning the loaded dispenser listed in Table I shall be conditioned at +145°F for at least 24 hours per MIL-STD-810, Method 501.

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4.3.2.2.4 Cold temperature conditioning. Following conditioning the loaded dispenser listed in Table I shall be conditioned at -35°F for at least 24 hours per MIL-STD-810, Method 502.

4.3.2.2.5 Ambient temperature conditioning. Following conditioning the loaded dispenser listed in Table I shall be conditioned at 70°F for at least 24 hours.

4.3.2.2.6 Function tests. The dispenser shall function when tested in the modes and temperatures noted above and paragraph 4.4.2.6.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48688A(AR)

PARAGRAPH	TITLE Dispenser and Mine, Ground: Anti-personnel/Antitank M131 System		SHEET 1 OF 1		DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Dispenser and Mine,Ground: AP/AT M 131 System (Dwg. 9337506)				
	Examination of defects	6		3.3	4.4.2.5
	Transportation Vibration (TV) test	6		3.3	4.3.2.2.1
	Seven Foot Drop test	6		3.3	4.3.2.2.2
	Hot temperature conditioning	2		3.3	4.3.2.2.3
	Cold temperature conditioning	2		3.3	4.3.2.2.4
	Ambient temperature conditioning	2		3.3	4.3.2.2.5
	Functioning	6		3.3	4.3.2.2.6 - 4.4.2.6

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

Maximum lot size shall be no more than one (1) month production (30 calendar day period).

4.4.2 Examination. See MIL-A-48078 and 4.4.2.1 through 4.4.2.6. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48688A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Coil and Spacer Assembly	1 OF 1	9337504
			NEXT HIGHER ASSEMBLY: 9337505

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Polarity of coil		100%	3.3	Test fixture
2	Correct connection of wires to coil		100%	3.3	Visual
Major					
101	Presence of parts: 1. Insulation sleeving 2. Flat cable 3. Solder sleeve 2 places 4. Spacer 5. Data transfer coil 6. Disk tape		0.40%	3.3	Visual
Minor					
201	Poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48688A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9337505
4.4.2.2	Loaded Tube Assembly M131 System				NEXT HIGHER ASSEMBLY 9337507
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Orientation of mine, at 3 places		100%	3.3	Visual during assembly
2	Orientation of coil & spacer assy 3 places		100%	3.3	Visual during assembly
Major					
101	Spring finger does not press directly on leadwire		0.40%	3.3	Visual during assembly
102	Presence of spider and orientation		0.40%	3.3	Visual during assembly
103	Presence of obturator and orientation		0.40%	3.3	Visual during assembly
104	Orientation of retainer plate assy 3 places		0.40%	3.3	Visual during assembly
105	Presence of conical spring 2 places and orientation		0.40%	3.3	Visual during assembly
106	Marking exists		0.40%	3.2	Visual during assembly
Minor					
201	Shear pin oriented symetrically		0.65%	3.3	Visual
202	Presence of caps, shear pin 2 places		0.65%	3.3	Visual
203	Presence of tape, pressure sensitive		0.65%	3.3	Visual during assembly
204	Mine assembly improper order in tube		0.65%	3.3	Visual
205	Presence of shear ring		0.65%	3.3	Visual
206	Poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48688A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.3	Loaded Module Assembly M131 System				9337507
					NEXT HIGHER ASSEMBLY 9337506
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Safe condition of out of line safety housing assembly		100%	3.3	Visual
Major					
101	Proper connection of leads, Note 3 of drawing		100%	3.3	Gage
102	Breech plug not fully seated		0.40%	3.3	Gage
103	Presence of parts: 1. Foamed module 2. Detonating cord assy 3. Cable and connector assy. 4. Housing assy 5. Tube assy 6. Breech spacer 7. Pressure seal disc 8. Cartridge assy. 9. Performed packing 10. Breech plug 11. Sealing compound 12. Initiation system disk		0.40%	3.3	Visual
104	Sealing compound missing from breech		0.40%	3.3	Visual
105	Solder workmanship		100%	3.3	3.3.1.3, 4.5, Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3 (cont.)	Loaded Module Assembly M131 System	2 OF 2	9337507
			NEXT HIGHER ASSEMBLY 9337506

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Presence of actuator and sealing compound		0.65%	3.3	Visual
202	Orientation of Cable and Connector Assy		0.65%	3.3	Visual
203	Cable secured by clamps		0.65%	3.3	Visual
204	Presence of parts		0.65%	3.3	Visual
205	Poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48688A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Plate Assembly, Retainer				9337511
					NEXT HIGHER ASSEMBLY 9337505
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Presence of stake		0.40%	3.3	Visual
102	Orientation of borerider pin		0.40%	3.3	Visual
103	Diameter of borerider pin		0.40%	3.3	Gage
104	Height of borerider pin		0.40%	3.3	Gage
Minor					
201	Poor workmanship		0.65%	3.7	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48688A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Dispenser and Mine, Ground: M131 (Mechanical)	1 OF 1	9337506
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Proper connection of MCD coil and cable and connector assy.		100%	3.3	Visual
Major					
101	Continuity test of connection in M131		100%	3.3	4.4.3.2.2,4.4.3.2.3, 4.4.3.2.4
102	Leak test		100%	3.3	4.4.3.2.5
103	CI self test		100%	3.3	4.4.3.3.2
104	CI data load test		100%	3.3	4.4.3.3.3
105	Presence of parts		100%	3.3	Visual
106	Marking improper		0.40%	3.3	Visual
107	Ramp test		100%	3.3	Visual
Minor					
201	Poor workmanship		0.65%	3.7	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Dispenser and Mine, Ground: M131 (Functional)	1 OF 1	9337506
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special A	Hazardous mine duds	1/	--	3.4	4.4.3
Major					
101	Deployment function	1/	--	3.3.2	4.4.3
102	Mine field pattern	1/	--	3.3.2	4.4.3
103	Mine function	1/	--	3.4	4.4.3/6.6

NOTES:
1/ See 4.4.3.1 for test quantities, test methods and acceptance/rejection criteria.

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4.4.3 Testing.

4.4.3.1 Testing of dispenser and mine, ground: antipersonnel/antitank M131. This section includes testing requirements that pertain to the dispenser and mine, ground: antipersonnel/antitank M131. Samples shall be tested in accordance with Table II.

QUALITY CONFORMANCE INSPECTION

PERFORMANCE SAMPLES
TABLE II

Dispenser Inspection Lot Size	Sample Size	Dispenser Command: Hard Wire 1 Hot	2 AMB	3 Cold	Rejection Inval Comm	R.F. 4 Hot	5 AMB	6 Cold
1-25	2	1	-	-	1	-	-	1
26-50	4	1	-	1	2	1	-	1
51-150	6	1	1	1	3	1	1	1
151-500	8	1	1	2	4	1	1	2
501-1200	10	2	1	2	5	2	1	2
Accept/Reject Criteria Sample Size	2	0-1	-	-	0-1	-	-	0-1
	4	0-1	-	0-1	0-1	0-1	-	0-1
	6	0-1	0-1	0-1	0-1	0-1	0-1	0-1
	8	0-1	0-1	0-1	0-1	0-1	0-1	0-1
	10	0-1	0-1	0-1	0-1	0-1	0-1	0-1

ACCEPTANCE/REJECTION CRITERIA BASED ON MINES

Sample	Mines Ejected	Deployment	Deployed Mines Function	Hazardous Duds
2	42	0-1	2-3	0-1
4	84	0-1	5-6	0-1
6	126	3-4	8-9	0-1
8	168	3-4	11-12	0-1
10	210	6-7	15-16	0-1

(The number of mines per dispenser is 21.)

NOTE: 1. Hard Wire - Hot - Coded - At .79 of first SD time envelope give RSD command with Signal Generator used to code dispenser and allow mines to function at RSD.
2. Hardwire - Amb. - Coded - At .79 of first SD time envelope give RSD command with Signal Generator used to code dispenser. At 1.50 of first SD time envelope give Command Destruct command.
3. Hard Wire - Cold - Coded (same as 1).
4. R.F. - Hot (same as 2).
5. R.F. - AMB (same as 1).
6. R.F. - Cold (same as 2).

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4.4.3.2 Safety determination and circuit integrity.

4.4.3.2.1 Out-of-line safety. Prior to assembly of the Housing Assembly, Out-Of-Line Safety Device (9292865) into the dispenser, the following conditions shall exists:

a. Interrupter- Present and in safe position

b. Spring- Present

c. Bellows Actuator- Present and unactuated with minimum gap between it and the interrupter.

4.4.3.2.2 Continuity/resistance of dispenser circuits. Prior to installation of the Control Indicator (Drawing 9333036), circuit continuity of the following circuits as measured from the connector cable and connector assembly (Drawing 9337525) shall meet the requirements listed (Reference wire diagram, MOPMS Dispenser Drawing 9337533).

Circuit	Connector Pin Number	Requirement
Tube 1	3 and 16	.275 to .350 ohms
Tube 2	4 and 17	.275 to .350 ohms
Tube 3	5 and 18	.275 to .350 ohms
Tube 4	6 and 19	.275 to .350 ohms
Tube 5	7 and 20	.275 to .350 ohms
Tube 6	8 and 21	.275 to .350 ohms
Tube 7	9 and 22	.275 to .350 ohms
Bellows Actuator	12 and 24	.200 to .270 ohms
Detonators	13 and 25	.500 to .900 ohms
MCD Coil	1 and 14	3.0 to 5.0 ohms

4.4.3.2.3 Magnetic coupling and device polarity. With a D.C. current of 75 ± 25 milliamperes applied to the connector of the dispenser cable and connector assembly (9337525) positive (+) lead on pin 14 and negative (-) on pin 1, a magnetic compass will point toward the MCD coil when it is brought near the coil on the coil center line.

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4.4.3.2.4 Insulation resistance test. The dielectric resistance between the shields, accessed through pin 10 of Connector cable and Connector Assembly 9337525, and the Control Indicator bracket of Loaded Modular Assembly 9337515 shall be 50 megohms or greater when measured with a 500 ± 50 volt megohmeter with a 2 microamps maximum.

4.4.3.2.5 Leak test. With an air pressure of .5 + .1 psi applied to the inside of the dispenser and allowed to stabalize, the pressure of the captive air shall not drop more than .05 psi in one (1) minute after the supply pressure has been shut off.

4.4.3.3 Dispenser function. Prior to any functional testing and with the Control Indicator knob in the "Position Safe", a power supply measuring no less than 14.4 volts (open circuit) will be installed into the control indicator of the dispenser. After assembly of this power supply, the dispenser will be conditioned as outlined in 4.4.3.5 and will function as follows:

4.4.3.3.1 Lamp check. The "Battery", "CKT OK" and "Data In" lamps must light when tested by pushing in on the lamp housing.

4.4.3.3.2 Self test. When the control indicator selector switch is turned to the "TEST" position, the "BATTERY" lamp on the control indicator face must light immediately, and the "CKT OK" lamp must light in less than 2 seconds. This test indicates that the battery voltage is adequate and that selected circuits within the control indicator and dispenser are functioning properly.

4.4.3.3.3 Data load. With the Control Indicator selector switch in the "Data Load" position, a signal generator properly located between the guide rails of the dispenser cover when properly activated to load data into the dispenser; the "DATA IN" light on the face of the control indicator shall light immediately indicating that a group identification codes has been loaded into the dispenser.

4.4.3.3.4 Arming of dispenser. The dispenser arming cycle begins when the selector switch on the control indicator is pushed in and turned to the "ARMED" position, and at 5 minutes and 35 seconds or later, the dispenser must accept a proper dispense command to eject the stored mines.

4.4.3.3.5 Dispense command - R.F. After arming per paragraph 4.4.3.3.4, the dispenser shall accept a properly coded "DISPENSE" command of 9.0 ± 1.0 MV/M field strength sent by a signal generator. "PROPERLY CODED" means the same code that was loaded into the dispenser in 4.4.3.3.3 was transmitted. If the dispenser was not "properly" coded, it must not accept a R.F. dispense command.

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4.4.3.3.6 Dispense command - hardwire. After arming per paragraph 4.4.3.3.4 the dispenser shall accept a "DISPENSE COMMAND" from a M32 or M34 blasting machine, or an M57 firing device when properly connected to the "HARDWIRE" terminals on the MCU.

4.4.3.3.7 Rejection of invalid R.F. command. After arming, paragraph 4.4.3.3.4, the dispenser shall reject any incorrectly coded, 9.0 MV/M or greater field strength signal from a Signal Generator.

4.4.3.3.8 Coding of mines. If a group identification code was loaded into the dispenser per paragraph 4.4.3.3.3 this identification code will be loaded into the mines after battery activation in the deployment cycle. Mines coded with a group identification code can have their self destruct time recycled or be "COMMAND DESTRUCTED" by the signal generator that coded the dispenser per 4.4.3.3.3.

If the dispenser did not receive a group identification code per paragraph 4.4.3.3.3 and received a dispense command, paragraph 4.4.3.3.5, a "BLANK" code will be loaded into mines after battery activation in the deployment cycle. Mines coded with a "BLANK" code cannot have their self destruct timer reset, nor "COMMAND DESTRUCTED" and they shall have only a single self destruct option 3.3.2.1.

4.4.3.3.9 Mine field pattern. The ejected mines shall fall in an area defined as a half section of an annulus with a minor radius of 10 meters and a major radius of 40 meters.

4.4.3.4 Mine function. After deployment, the mines when the proper coding shall be capable of functioning in the prime target sensor mode, a self-destruct mode, recycled self-destruct mode and command destruct mode.

4.4.3.4.1 Self-destruct failure. The armed mine which fails to self destruct after the maximum (SD) time as specified on 9333026 has elapsed, shall not be capable fo being functioned by trip line sensor target sensor, command destruct or any other stimuli.

4.4.3.4.2 Recycled self-destruct (RSD) function. The self-destruct time out function of an armed and unevented mine shall be capable of being reset or recycled if the mine received a group identification code. With a group identification code in the mine, the self-destruct time out function shall be reset to "zero" when the mine receives a RSD command from the signal generator, used to code the deploying dispenser, at a R.F. field strength of 9.0 + 1.0 MV/M. Once reset, the mine shall function within the self destruct time measured from the time of the RSD command.

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4.4.3.4.3 Command destruct. With a group identification code in the mine, the armed and unevented mine shall be capable of being functioned when it receives the Command Destruct command from the signal generator, used to code the deploying dispenser, at a R.F. field strength of 9.0 + 1.0 MV/M.

4.4.3.5 Lot acceptance testing. The dispenser lot acceptance test shall be subjected to conditioning and testing as outlined in 4.3.2.2.3, 4.3.2.2.4, 4.3.2.2.5, 4.3.2.2.6 and 4.4.2.6.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit inspection equipment designs to the Government for approval in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and 6.3 herein.

4.5 Soldering process control plan. The contractor shall have a soldering process control plan for use in conjunction with high reliability soldering system. The soldering process control plan shall comply with all of the specified requirements noted in MIL-S-45743 and MIL-S-46844. Documentation control shall be in accordance with MIl-Q-9858. The soldering process control plan shall be prepared under the direction of personnel with Category C Authority.

In cases where a connection is not shown in the applicable specifications or drawings, the contractor shall highlight in his document a description of the aformentioned solder connection.

In addition, the soldering process control document shall indicate methods of training, length of training and category of personnel utilized in this soldering operation.

The contractor's soldering process control document shall be approved by the technical agency prior to first article testing or start of production. (Data Item DI-OT-88-12199 shall apply)

5. PACKAGING

5.1. Preservation and packaging.

5.1.1. Level A. Preservation and packaging shall be in accordance with Dwg. 9349988.

5.2. Packing.

5.2.1. Level A. Packing shall be in accordance with Dwg. 9349988.

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5.3. Marking.

5.3.1. Level A. Marking shall be in accordance with Dwg. 9349988.

6. NOTES

6.1 Intended use. The Mine, Antitank/Antipersonnel: High Explosives covered by this specification is intended for use as an antitank and antipersonnel mine dispensed from the MOPMS Dispenser.

6.2 Acquisition requirements. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, AMCCOM, ATTN: AMSMC-QAT-I(D), Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Data submissions. Results of all tests performed by the contractor shall be submitted in accordance with Data Item DI-R-1721 on the DD Form 1423 for the contract. Distribution shall include the following:

Commander
ARDEC
ATTN: AMSMC-QAT-M (D)
Picatinny Arsenal, NJ 07806-5000

6.5 Distribution of ammunition data cards and test data. Distribution of data cards shall include the following: Commander, U.S. Army Armament Research Development and Engineering Center, AMSMC-QAT-M(D), Picatinny Arsenal, New Jersey 07806-5000.

6.6 Drawings. Drawings in Section 2 of this specification under the heading US Army Armament Research Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under cognizance of ARDEC.

6.7 Definitions. For purposes of the tests in this specification, the following terms are defined:

a. Premature arming. Is defined as arming of the S&A assy (evidenced by actuation of piston actuator) before the minimum arming actuator output time specified on Drawing 9333026.

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b. Premature detonation. Is defined as detonation of a mine (S&A armed) occurring before the minimum sensor enable time specified on Drawing 9333026.

c. Early self destruct. Is defined as detonation of the mine (S&A armed) occurring at or after the minimum sensor enable time specified on Drawing 9333026.

d. Enable time. If defined as the maximum sensor enable time specified on Drawing 9333026, plus 5 minutes, after application of the electromagnetic activation pulse.

e. Hazardous dud. Is defined as an armed mine capable of being functioned, after the Self Destruct time has elapsed, by target sensor, self-destruction or any other stimuli.(See 3.3.2)

f. Non-hazardous dud (Dud). Is defined as a mine incapable of being functioned, after the set SD time has elapsed, by any of the modes listed in 6.6e above. (See 3.3.2)

g. High order functioning. HE Mine: High order functioning will be evidenced by high order detonation of the Main charge train.

6.8 Subject term (key word) listing.

Dispenser, Mine

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345-A544)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-48688A (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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